

Inductive Learning

Session Outcomes

- Inductive Learning
- Hypothesis Space
- Dataset Spilt
- Cross validation

- Machine learning is given a set of training examples:

$$(\hat{x}, y)$$

- $\hat{x}$ represents the input feature vector.
- y represents the target or output attribute.
- The feature vector consists of multiple features:

$$\hat{x} = (x_1, x_2, \dots, x_n)$$

- The goal is to learn a function: $f(\hat{x})$ such that

$$y = f(\hat{x})$$

Learning Objective

- Learn a function

$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

- Such that

$$y = f(\mathbf{x})$$

- The learned function should correctly predict the output for unseen inputs.

Types of Learning Problems

- **Classification**

$$f(\mathbf{x}) \in \{c_1, c_2, \dots, c_k\}$$

- Output is discrete.
- Predicts one among a finite set of classes.

- **Regression**

$$f(\mathbf{x}) \in \mathbb{R}$$

- Output is continuous.
- Predicts a numerical value.

- **Probability Estimation**

$$f(\mathbf{x}) = P(y \mid \mathbf{x})$$

- Estimates the probability of an outcome.

Inductive Learning

- Given training examples

$$D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)\}$$

- Learn a function

$$f \in H$$

where H is the hypothesis space.

- The learning algorithm searches for a function that best explains the observed data.
- This process of deriving a general rule from specific examples is called **Induction**.

$$D \rightarrow \text{Learning Algorithm} \rightarrow f(\mathbf{x})$$

Hence, classification, regression, and probability estimation are collectively known as inductive learning problems

Hypothesis

- In machine learning, the true relationship between input and output is represented by an unknown function:

$$y = f(\mathbf{x})$$

- Machine learning attempts to learn an approximation of this function.
- The learned function is called a **Hypothesis** and is denoted by:

$$h(\mathbf{x})$$

- Thus,

$$h(\mathbf{x}) \approx f(\mathbf{x})$$

Hypothesis Space

- A learning algorithm does not consider only one hypothesis.
- It searches through a collection of possible hypotheses.
- The set of all candidate hypotheses is called the **Hypothesis Space**.
- It is denoted by:

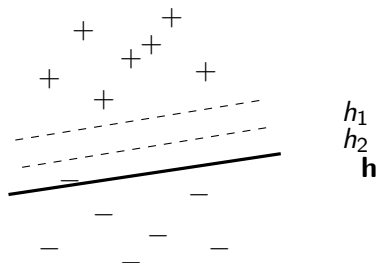
$$H = \{h_1, h_2, h_3, \dots\}$$

- The learning algorithm searches the hypothesis space and selects the best hypothesis.
- Therefore,

$$h \in H$$

Training Data $\rightarrow$ Learning Algorithm $\rightarrow h$

Classification Problem and Hypothesis Space



- Each line represents a possible hypothesis.
- All possible hypotheses form the hypothesis space H .
- The learning algorithm selects the best hypothesis:

$$h \in H$$

Testing Machine Learning Algorithms

- The goal of machine learning is to make accurate predictions.
- Performance cannot be judged only using the training data.
- A model should perform well on unseen data.
- Therefore, machine learning models must be evaluated using separate datasets.

Training Data $\rightarrow$ Learn Model $\rightarrow$ Test on New Data

Overfitting

- A model should learn the underlying pattern in the data.
- If training continues excessively, the model may memorize the training examples.
- This phenomenon is called **Overfitting**.
- An overfitted model performs well on training data but poorly on unseen data.

Desired Behaviour

Learn Pattern $\neq$ Memorize Data

Training, Validation and Test Sets

$$\text{Dataset} \rightarrow \begin{cases} \text{Training Set} \\ \text{Test Set} \end{cases}$$

- **Training Set**

- Train the model.
- Learn patterns from data.

- **Test Set**

- Evaluate the final model.
- Measure performance on unseen data.
- Training and test sets should not overlap.

$$\text{Training Set} \cap \text{Test Set} = \emptyset$$

- This ensures an unbiased estimate of model performance on unseen data.

Training and Validation Split

$$\text{Training Set} \rightarrow \begin{cases} \text{Train} \\ \text{Validation} \end{cases}$$

- **Train**

- Learn model parameters.

- **Validation**

- Hyperparameter tuning.
- Model selection.

- Train and Validation sets should be independent:

$$\text{Train} \cap \text{Validation} = \emptyset$$

The sets should not share examples

Training and Validation split

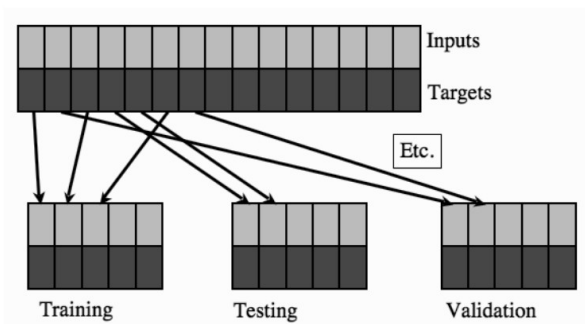


Figure: 50:25:25 (large Data) 60:20:20 (Average Data)

Hypothesis Space and Model Selection

Training Data

Data	x	y
Training	1	2
Training	2	4
Validation	3	6
Testing	4	8

The learning algorithm searches for a function that explains the data.

$$H = \{h_1, h_2, h_3, \dots\}$$

where

- h_1 : Degree 1 polynomial
- h_2 : Degree 2 polynomial
- h_3 : Degree 3 polynomial

The set of all candidate models is called the **Hypothesis Space**.

Hypothesis Space and Model Selection

The set of all candidate models considered by a learning algorithm is called the **Hypothesis Space**.

$$H = \{y = ax + b, y = ax^2 + bx + c, y = ax^3 + bx^2 + cx + d, \dots\}$$

Each candidate model is called a **Hypothesis**. **Training Data Validation**

Example:

$$x = 3, \quad y = 6$$

Hypothesis Space

$$H = \{h_1, h_2, h_3\}$$

Hypothesis	Model
h_1	$y = 2x$
h_2	$y = x^2 - x + 2$
h_3	$y = x^3 - 2x^2 + 3x$

Validation Error for Different Hypotheses

Validation Point: $(x, y) = (3, 6)$

Hypothesis	Prediction ($\hat{y}$)	Error Formula	Error
h_1	6	$(6 - 6)^2$	0
h_2	8	$(6 - 8)^2$	4
h_3	18	$(6 - 18)^2$	144

Observation

- h_1 produces the smallest validation error.
- h_2 and h_3 are more complex models.
- Increasing model complexity does not necessarily improve performance.
- The validation set helps us choose the best hypothesis.

Selected Hypothesis = $h_1(x) = 2x$

Testing the Selected Hypothesis

Selected model:

$$h_1(x) = 2x$$

Test point:

$$(x, y) = (4, 8)$$

Prediction:

$$\hat{y} = h_1(4) = 2(4) = 8$$

Test Error:

$$(8 - 8)^2 = 0$$

Test Error = 0

Discussion

- A hypothesis is a candidate model that can explain the data.
- The collection of all candidate models is called the **Hypothesis Space**.

$$H = \{h_1, h_2, h_3, \dots\}$$

- The learning algorithm searches the hypothesis space.
- Each hypothesis is evaluated using the validation data.
- The hypothesis with the lowest validation error is selected.
- In this example,

$$h_1(x) = 2x$$

gives the smallest validation error and is therefore chosen as the final hypothesis.

- Since the test error is also zero, the chosen model generalizes well.

$$h_1 \in H$$

Disadvantages of Validation Approach

- Validation approach is sensitive to the specific split used.
- Cross-validation reduces variance by testing across multiple splits.
- It ensures all samples are used in validation at least once.

Cross Validation

- Sometimes the available data is limited.
- The dataset is divided into K equal parts.
- One part is used for validation while the remaining parts are used for training.
- The process is repeated K times.

Types of Cross Validation

Cross Validation is used to obtain a reliable estimate of model performance.

Common techniques are:

- 1 Leave-One-Out Cross Validation (LOOCV)
- 2 Leave-p-Out Cross Validation (LpOCV)
- 3 K-Fold Cross Validation
- 4 Stratified K-Fold Cross Validation

Leave-One-Out Cross Validation (LOOCV)

Suppose the dataset contains 500 records.

For each experiment:

- One sample is used for validation.
- Remaining $n - 1$ samples are used for training.

Experiment 1 : 1 Validation + 499 Training

Experiment 2 : 1 Validation + 499 Training

⋮

Experiment 500: 1 Validation + 499 Training Final Accuracy

$$= \frac{ACC_1 + ACC_2 + \cdots + ACC_{500}}{500}$$

Validation data differs at each training

Advantages and Disadvantages of LOOCV

Advantages

- Almost all data is used for training.
- Every sample is used for validation once.

Disadvantages

- Only one record is used for validation.
- Computational cost is very high.
- Model must be trained n times.

Leave-p-Out Cross Validation

Instead of leaving one sample out, we leave p samples out.

Examples:

$$p = 10, \quad p = 20, \quad p = 30$$

Procedure:

- Select p samples for validation.
- Use remaining samples for training.
- Repeat for different combinations.

Final performance is obtained by averaging all validation results.

K-Fold Cross Validation

Assume

$$n = 500, \quad k = 5$$

Validation size:

$$\frac{500}{5} = 100$$

The dataset is divided into 5 equal folds.

- Fold 1 used for validation
- Remaining folds used for training
- Repeat until every fold becomes validation once

K-Fold Cross Validation Example

Experiment	Validation Fold
1	Fold 1
2	Fold 2
3	Fold 3
4	Fold 4
5	Fold 5

Final Accuracy

$$\text{Accuracy} = \frac{\text{ACC}_1 + \text{ACC}_2 + \text{ACC}_3 + \text{ACC}_4 + \text{ACC}_5}{5}$$

For classification problems:

- Class distribution may not be preserved.
- One fold may contain more samples from one class.
- Another fold may contain fewer samples from that class.

This may lead to unreliable validation results.

Stratified K-Fold Cross Validation

Stratified K-Fold preserves the class distribution in every fold. Example:

$n=500$, $k=5$

Each fold contains samples of:

$$\frac{500}{5} = 100$$

If the original dataset contains 60% Class A and 40% Class B then every fold approximately contains 60% Class A and 40% Class B